

Distributed MAC Scheduling Scheme for C-RAN with Non-Ideal Fronthaul in 5G Networks

Min Huang, Xu Zhang

Communication Infrastructure Research Lab, Intel Labs China
email: min.a.huang@intel.com

Abstract—A typical cloud-radio access network (C-RAN) architecture in fifth-generation (5G) network is composed of one radio cloud center (RCC), moderate amount of radio aggregation units (RAU) and massive remote radio heads (RRH). This paper proposes a distributed medium access control (MAC) scheduling scheme for C-RAN, where RCC and RAU are connected with non-ideal fronthaul. In this scheme, a full scheduling operation and a limited scheduling operation are executed at RCC and RAU, respectively. The full scheduling operation at RCC is based on the overall network information and hence could generate a global-optimized scheduling result with ideal fronthaul. The limited scheduling operation at RAU is based on the latest hybrid automatic repeat request (HARQ) feedback information and channel state information (CSI), and could provide real-time supplement when the fronthaul transfer between RCC and RAU is delayed or lost. The integration of these two scheduling results are also provided to further exert their individual advantages. Simulation results demonstrate that the proposed distributed MAC scheduling scheme could effectively mitigate the performance loss due to non-ideal fronthaul in terms of cell sum throughput and cell edge throughput, and therefore benefits in promoting the feasibility of C-RAN in 5G network architecture evolution.

Index Terms—5G, cloud-radio access networks (C-RAN), medium access control (MAC) scheduling, fronthaul, distributed.

I. INTRODUCTION

Cloud-radio access network (C-RAN) is recognized as one of the advantageous architectures for next generation wireless communication network i.e. fifth generation (5G) cellular mobile communication system [1], due to its capability to centralize processing resource, perform large-scale coordination, reduce energy consumption and enable elastic network structure [2]-[5]. A typical network architecture for C-RAN consists of one radio cloud center (RCC), moderate amount of radio aggregation units (RAU) and massive remote radio heads (RRH), which are connected by so-called fronthaul interface [6][7].

In the architecture of fronthaul-based C-RAN, RAN function split between RCC and RAU is hotly discussed [8]-[10]. One of the popular modes is to place medium access control (MAC) layer function in RCC and physical (PHY) layer function in RAU. The advantage of such split lie in a) the traffic volume in fronthaul could be greatly reduced than the way of also placing a part of PHY function at RCC, b) the fast dynamic inter-cell coordination algorithm, like coordinated scheduling (CS) or coordinated beamforming (CB), could be

possible at different RAUs with the common central MAC controller at RCC, thus the multi-cell throughput could be improved than the way of placing all MAC functions at RAU. In this mode, the PHY-MAC interface messages are transferred between RCC and RAU via fronthaul interface.

However, in practical C-RAN operations with MAC-PHY function split, the non-ideal fronthaul message transfer, e.g. the possible large latency and jitter encountered by the packets transferred in the fronthaul interface, turns out to be a challenging problem to block C-RAN commercial deployment. To meet the service latency request and hybrid automatic repeat request (HARQ) round-trip time (RTT) in long-term evolution (LTE) and 5G standards, the packet transfer latency in fronthaul interface should be restricted within a fraction of one mini-second. Otherwise, the following RAN protocol processing are severely impacted without sufficient input information, such as the scheduling grant, the HARQ feedback, the channel state information (CSI), and the quality-of-service (QoS) information.

Several approaches are provided by literatures to cope with this problem. One of them is to prioritize the traffic with tight latency budget in fronthaul traffic arrangement and optimize the system based on these restrictions. In [11], the resource allocation and beamforming design for system utility maximization under given fronthaul rate constraints and QoS requirements at users are considered, and a quasi-geometric programming problem is formulated to reduce the processing time. Another approach is to compress the data in fronthaul to lower down its traffic. In [12], the partial bit sampling for 50% fronthaul data compression is proposed and experimentally implemented in field programmable gate array (FPGA). In [13][14], a joint quantization mapping across all fronthaul links is implemented which is adapted with the current channel conditions and could yield significant performance gain over per-fronthaul link point-to-point quantization. Although by these approaches, the occurrence rate of fronthaul transfer failure could be decreased in a certain degree, the residual packet delay or lost would still severely impair C-RAN practical performance.

In such background, this paper walks along another road to cope with the fronthaul non-ideality on the root. Specifically, it proposes a distributed MAC scheduling scheme for C-RAN architecture, in which a full scheduling operation and a limited scheduling operation are executed at RCC and RAU

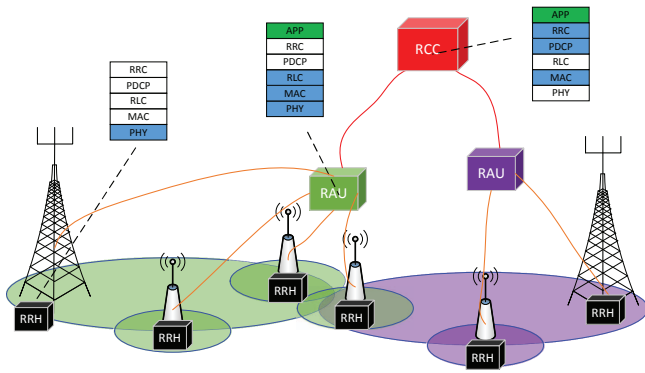


Fig. 1. C-RAN Architecture and RAN function split.

respectively. The full scheduling operation at RCC is based on the overall network information including service requirement, resource utilization, channel state and interference situation of all involved cells, and hence it is able to generate a global-optimized scheduling result when the fronthaul message transfer works normal. The limited scheduling operation at RAU provides real-time supplement with that at RCC when the fronthaul message transfer is delayed or lost, so that the performance degradation due to such fronthaul accidents is prevented or compensated. Even though the fronthaul message transfer is punctual, the integration of the scheduling results from these two sources is also beneficial, which could exert their individual advantages. To disclose the performance of such distributed MAC scheduling scheme, a system-level simulator with massive multi-input multi-output (MIMO) and 5G frame structure is built, and the metrics including cell sum throughput and cell edge throughput are evaluated.

The remainder of this paper is organized as follows. Section II introduces the C-RAN architecture and RAN protocol split. Section III formulates the problem caused by non-ideal fronthaul, and then illustrates the proposed distributed MAC scheduling scheme. Section IV presents simulation results to exhibit the performance and reveal the benefits. Section V draws the conclusions.

II. C-RAN ARCHITECTURE FOR 5G SERVICES

An exemplified model of C-RAN architecture including one RCC, two RAUs and six RRHs is shown in Fig. 1. The RCC overall controls the whole network, while the green RAU and purple RAU control the RRHs in left-half area and the ones in right-half area respectively, with a certain part of their coverages overlapped.

A bundle of RAN protocol function split is given in [9][10][15], and this paper concentrates on the PHY-MAC split mode. On the original idea of this mode, PHY function lies in the distributed RRHs while MAC and above functions lie in central base-band unit (BBU). This paper extends this ideas in two aspects. First, RRH only executes PHY pre-processing, but the rest main body of PHY processing is executed in RAU. RAU has small-scope centralization capability covering a local heterogeneous network (HetNet) by which the technique of

TABLE I
RAN PROTOCOL FUNCTION SPLIT FOR RRH, RAU AND RCC.

	User-plane Functions	Control-plane Functions	Flexibility	Hardware platform
RRH	PHY pre-processing	N/A	low	Dedicated RF chip with certain baseband processing capability
RAU	PHY processing, MAC/RLC packet handling, APP	limited MAC control	medium	DSP chip, FPGA, accelerator for specific functions
RCC	PDCP, APP	full MAC control, RRC handling, RRM, SON	high	GPP

joint transmission (JT) or joint reception (JR) belonging to coordinated multi-point (CoMP) could be enabled [16]. Second, one part of MAC control function is taken by the RAU, while the other part of the MAC control function is taken by RCC which has large-scope centralization and global optimization capabilities. The whole RAN protocol layer function splits among RCC, RAU and RRH are summarized in Table I¹.

The purposes of this C-RAN architecture are as follows:

- Placing PHY pre-processing at RRH is to convert the physical antenna ports to virtual antenna ports for massive MIMO [17]. Placing the main body of PHY at RAU is to enable CoMP JT and JR in intra-RAU cells.
- Placing the main body of MAC control at RCC is to enable CoMP with CS and CB in intra/inter-RAU cells. The reason to place some limited MAC control at RAU would be explained in the following sections.
- Placing RLC at RAU is to facilitate carrier aggregation (CA) involving the intra-RAU cells. If the fronthaul transfer between two RAUs is sufficiently swift, like within one transmission time interval (TTI), the CA function could also be made within inter-RAU cells.
- Placing PDCP at RCC is to facilitate dual/multi-connectivity within the whole HetNet, regardless of the type and amount of connected RAUs.
- APP server could be placed in RAU and RCC, providing quick-reaction and high-throughput mobile edge computing (MEC) services.

Regarding the connections among RRH, RAU and RCC,

- Between RAU and RRH: it mainly contains I/Q data for virtual antenna ports and control information for antenna combination. Its data rate is 1Gbps per port.
- Between RCC and RAU: it mainly contains MAC scheduling signaling, PDCP packet and RRC control

¹Abbreviations: radio resource management (RRM), self-optimization network (SON), digital signal processing (DSP), field programmable gate array (FPGA), general-based processor (GPP), packet data convergence protocol (PDCP), radio link control (RLC), application (APP), radio frequency (RF)

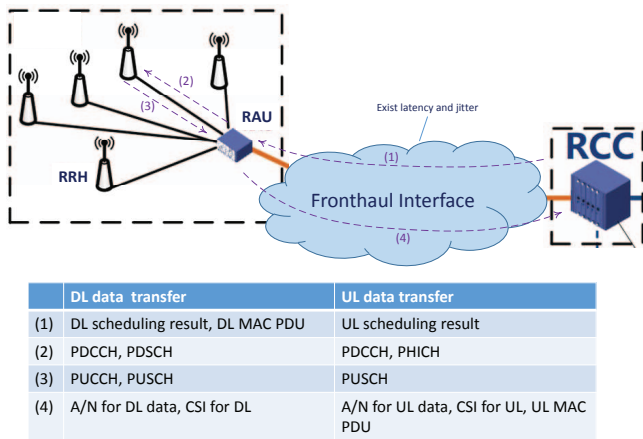


Fig. 2. Block diagram and message transfer in the fronthaul-based C-RAN architecture with the conventional scheme.

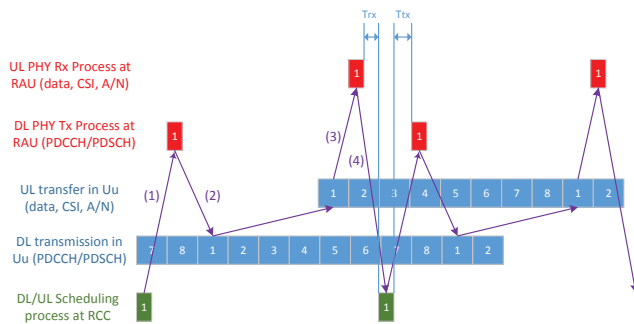


Fig. 3. Process flow diagrams and function split in the fronthaul-based C-RAN architecture with the conventional scheme. The digits in blocks represent the indexes of each HARQ process, and process 1 is exemplified for illustration.

signaling. Its data rate is 1Mbps per cell for control-plane and 10Gbps per cell for user-plane. Since control-plane is more critical for system maintenance, the fronthaul packet of control-plane normally has higher priority than that of user-plane in fronthaul packet switching/routing.

III. DISTRIBUTED MAC SCHEDULING FOR C-RAN

A. Problem Formulation

Fig. 2 and Fig. 3 show the function split and message signaling for downlink (DL)/uplink (UL) data transfer within the conventional C-RAN architecture. Three operations are involved in the whole process, therein DL/UL scheduling processing is executed at RCC, while UL PHY Reception (Rx) processing (including PUCCH and PUSCH) and DL PHY Transmission (Tx) processing (including PDCCH, PDSCH, PHICH) are executed at RAU.² Four messages are given in Fig. 2, where message 1 and 4 are transferred in fronthaul interface between RCC and RAU, and message 2 and 3 are transferred between RAU and RRH. The interval between UL PHY Rx processing at RAU and DL/UL Scheduling processing at RCC (denoted as T_{rx}) and the interval between DL/UL

Scheduling Process at RCC and DL PHY Tx processing at RAU (denoted as T_{tx}) are marked in Fig. 3.

Due to the delay or loss of message transfer in fronthaul, the operation and performance of C-RAN architecture would be suffered, as in the following use cases:

Use case 1: T_{tx} exceeds the restricted threshold or message 1 is lost, and hence scheduling result by RCC doesn't arrive at RAU before the start of DL PHY Tx processing. RAU cannot prepare PDCCH/PDSCH for the corresponding subframe, so that both new transmission and retransmission cannot be sent. Cell throughput is suffered by the interrupted data transfer.

Use case 2: For DL data transfer, T_{rx} exceeds the restricted thresholds or message 4 is lost, and hence scheduling result from RCC to RAU doesn't involve DL retransmission grant, or the involved DL retransmission grant doesn't come from the correct user equipment (UE) feedback. Since DL HARQ is asynchronous in LTE standard, it is allowed to involve the DL retransmission in the scheduling result of next subframe, which causes the transfer latency for the data packet of the corresponding HARQ process slightly increased. If the involved DL retransmission grant come from an incorrect HARQ feedback, the indicated DL data transfer is excessive and hence cell throughput is suffered.

Use case 3: For UL data transfer, T_{rx} exceeds the restricted thresholds or message 4 is lost, and hence scheduling result from RCC to RAU doesn't involve UL retransmission grant, or the involved UL retransmission grant doesn't come from the correct ACK/NACK (A/N) result. If scheduling result from RCC to RAU doesn't involve UL retransmission grant, the synchronous HARQ retransmission would fail and this HARQ process has to wait for next RTT instance, which causes the data packet transfer latency increase and decoding performance degrade. This has negative impact on user experience, especially for those time-critical services, like voice-over-IP (VoIP).

Use case 4: T_{rx} exceeds the restricted thresholds or message 4 is lost, which results in the latest CSI are absent in the RCC scheduling, and hence causes over- or under-estimated channel states and improper link adaptation result.

B. Proposed Scheme with Distributed MAC Scheduling

To cope with the message transfer delay or lost in fronthaul interface, a distributed MAC scheduling scheme is proposed as follows:

- A *full* scheduling function is placed at RCC and a *limited* scheduling function is placed at RAU; RCC transfers the scheduling result to RAU.
- If RAU receives the scheduling result by RCC prior to the start of DL PHY Tx processing, it would integrate its self-made scheduling result with the received scheduling result;
- Otherwise, if RAU doesn't receive the scheduling result by RCC prior to the start of DL PHY Tx processing, it would use its self-made scheduling result solely.
- RAU indicates the final scheduling result to RCC, if it is not identical to the scheduling result by RCC.

²The abbreviations for the physical channels are found in [18].

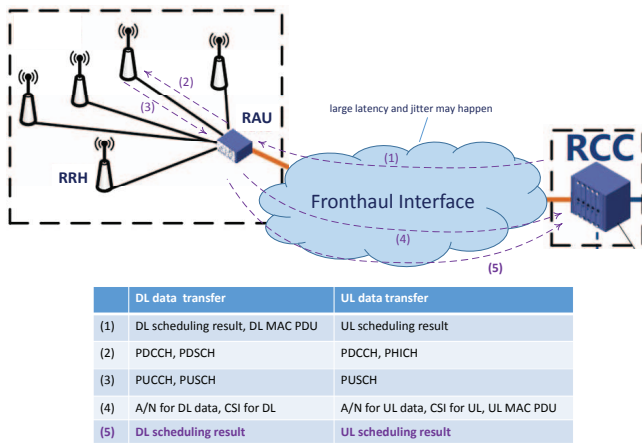


Fig. 4. Block diagram and message transfer in the fronthaul-based C-RAN architecture with the proposed scheme.

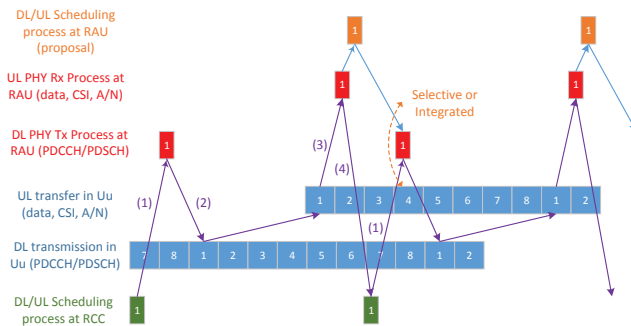


Fig. 5. Process flow diagrams and function split in the fronthaul-based C-RAN architecture with the proposed scheme. The digits in blocks represent the indexes of each HARQ process, and process 1 is exemplified for illustration.

Fig. 4 and Fig. 5 show the whole process of the proposed scheme. In each subframe and for both DL and UL data transfer, RCC makes the scheduling in a full mode, and sends the scheduling result to RAU (message 1). Here the scheduling operation includes selecting the UEs and determining the assigned time-frequency-space resource and transport format for each of them. In parallel, RAU makes the scheduling in a limited mode. RAU uses the scheduling result by RCC if it is received before the start of DL PHY Tx processing, in the way of making integration with its own scheduling result; otherwise, RAU uses its own scheduling result.

In Fig. 4, message 5 is added by the proposed scheme, which is used to indicate to RCC about the scheduling result by RAU. With such information, RCC could update its corresponding contexts and improve the scheduling for the next subframes, e.g. record the new transmission and retransmission scheduled by the RAU, update the weights for the scheduled UEs, etc.

In Fig. 5, the operation of DL/UL scheduling process at RAU is also added by the proposed scheme. Since it is executed at RAU, there is no interval with the UL PHY Rx processing ahead or the following DL PHY Tx processing,

TABLE II
THE CHARACTERISTICS OF SCHEDULING RESULTS BY RCC AND BY RAU.

Scheduling result	Pros	Cons
by RCC	owns overall information for global optimization among all cells; support dynamic inter-cell coordination and CoMP, especially benefit in cell-edge UE	sometimes lacks of the latest or correct A/N or CSI information
by RAU	always has the latest and correct A/N and CSI information	owns information about partial cells, supports semi-static inter-cell coordination

TABLE III
TYPICAL APPROACHES OF FULL SCHEDULING BY RCC AND LIMITED SCHEDULING BY RAU.

Approach index	Full scheduling by RCC	Limited scheduling by RAU
1	both new transmission and retransmission	only retransmission
2	for all connected UEs	for only cell-center UEs
3	with all resource blocks	with partial resource blocks
4	with all beamforming codes	with partial beamforming codes
5	for services with all QoS classes	for time-critical services

so the RTT period can be completely utilized for scheduling processing.

To reduce energy consumption at RAU, the scheduling at RAU could be activated or deactivated on a necessary basis. When the fronthaul network is lightly loaded, the happening possibility of over-length latency for message transfer (i.e. $T_{tx} + T_{rx}$ is larger than a restricted threshold) is low, then the RAU scheduling function is deactivated; on the opposite, when the fronthaul network is heavily loaded, the happening possibility of over-length latency for message transfer is high, then the RAU scheduling function is re-activated.

C. Full Scheduling vs. Limited Scheduling

The comparison between the scheduling results by RCC and RAU is shown in Table II. If the A/N and CSI information from RAU arrive at RCC punctually, the scheduling by RCC could maximally obtain the inter-cell coordination gain and hence lead to the optimal performance. On the opposite, if these feedback information or the scheduling result by RCC are delayed or lost, the scheduling in RAU could compensate the lost with sub-optimal solution.

Because of these characteristics, five typical approaches of full scheduling by RCC and limited scheduling by RAU are given in Table III:

- The purpose for Approach 1 is to cut down the scheduling burden by RAU. It only retains the necessary HARQ retransmission, in order to maintain HARQ process normal operation.
- The purpose for Approach 2 to 4 is to reduce the possibility that the scheduled UEs are interfered severely by neighbor cells or cause severe interference to neighbor

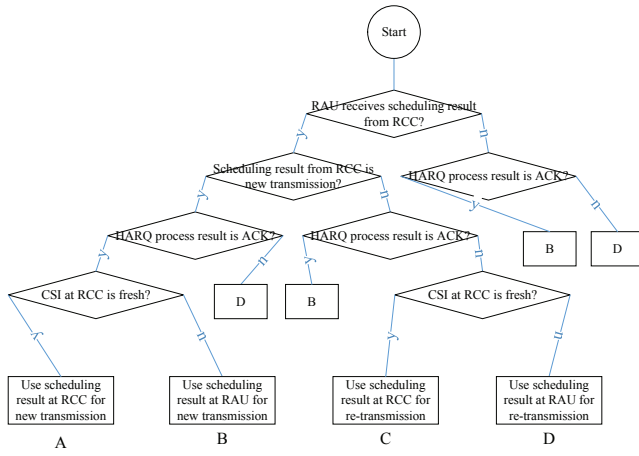


Fig. 6. Process flow diagram of scheduling results integration at RAU.

cells in the limited scheduling result, when the inter-cell coordination is impossible. In Approach 2, the signal to or from the cell-center UE is less likely to cause interference to or suffer interference from the neighbor cells. In Approach 3 and 4, the partial resource blocks or partial beamforming weights in limited scheduling could help enable the semi-static inter-cell coordination methods.

- The purpose for Approach 5 is to meet the stringent latency requirement for e.g. 5G ultra-robust low-latency communication (URLLC) services, like vehicle-to-everything (V2X). In this case, a certain portion of physical or logical radio resource is pre-allocated as a pool for the scheduling at RAU.

D. Scheduling Results Integration

When the scheduling results by RCC and by RAU are both available, their integration could bring additional gain over solely adopting RCC scheduling result. This integration could be based on HARQ A/N information, CSI or the mixture of these two. Fig. 6 presents the process flow diagram of the integration at RAU.

a) Based on HARQ A/N Information. RAU checks the scheduling result from RCC: if the A/N result of corresponding HARQ process is NACK but the scheduling result by RCC doesn't involve the retransmission grant, as Use Case 3 in Section III.A, RAU inserts or substitutes with its own scheduling result for the HARQ retransmission into the scheduling result by RCC; if the A/N result of corresponding HARQ process is ACK but the scheduling result by RCC involves the retransmission grant, as Use Case 2 and 3 in Section III.A, RAU removes the scheduling result for the HARQ retransmission from the scheduling result by RCC, or substitutes with its own limited scheduling result.

b) Based on CSI. When the scheduling results by RCC and by RAU are both available and the condition of above HARQ feedback factor is not met, the freshness of the used CSI at RCC dominates the further integration. The RCC indicates

TABLE IV
TYPICAL APPROACHES OF *Full Scheduling* BY RCC AND *Limited Scheduling* BY RAU.

Type	Value
Cell deployment	19 base stations with 57 cells, inter-site distance (ISD) is 500 meters
UE number and distribution	100 UEs are distributed randomly, RSRP range is [-110dBm, -70dBm]
Channel model	Follow [19], 3D urban macro, NLOS, sub-6 GHz carrier frequency
UE moving speed	30km/h
Frame structure	self-contained subframe, time division duplex (TDD) [20]
Bandwidth	20MHz
eNB antenna	64-antenna panel with 8 rows and 4 columns of cross-polarized elements
UE antenna	UL: 1 Tx antenna; DL: 2 Rx uncorrelated antennas
eNB Tx Power	46 dBm
UE maximum Tx power	23 dBm
UL targeted receiving power (P0)	-80 dBm

the freshness (such as the reception time) of its utilized CSI to RAU associated with the scheduling grant, and then RAU would check: if such indicated CSI is sufficiently fresh, the scheduling result by RCC is selected; otherwise, if it is insufficiently fresh, e.g., it is obsolete due to missing the latest CSI report, the improper scheduling result by RCC is replaced by the scheduling result by RAU for the corresponding UE.

IV. SIMULATION RESULTS

A system-level simulator is built to reveal the performance benefit of the proposed distributed MAC scheduling scheme, whose scenario and parameters are listed in Table IV. Therein, the latest Verizon 5G frame structure is adopted [20]. Besides, the technique of massive MIMO with 64 antennas at RRH is activated, by which multiple UEs are scheduled to share at most 48 spatial streams for DL and 24 spatial streams for UL at each resource block.

The cell sum throughput and cell edge throughput (10%-tile UE) are evaluated for four cases: 1) single cell without inter-cell interference (ICI); 2) multi-cell with ICI, C-RAN utilized, conventional centralized MAC scheduling only at RCC; 3) multi-cell with ICI, C-RAN utilized, proposed distributed MAC scheduling at both RCC and RAU; 4) multi-cell with ICI, C-RAN not utilized, independent scheduling at each RAU. For the proposed scheme, CoMP technique (including JT and JR for the cells controlled by a common RAU, CS and CB for the cells controlled by a common RCC) is activated in the full scheduling at RCC, while partial beamforming weights are utilized in the limited scheduling at RAU. The scheduling results integration follows Fig. 6. The performance curves are drawn when the fronthaul message loss rate between RCC and RAU is increased from 0 to 20%.

From the results in Fig. 7 and Fig. 8, when the loss rate is zero, i.e. with ideal fronthaul, the single cell (case 1) could achieve DL 2.76 Gbps and UL 1.74 Gbps cell sum throughput and DL 5.26 Mbps and UL 5.73 Mbps cell edge throughput

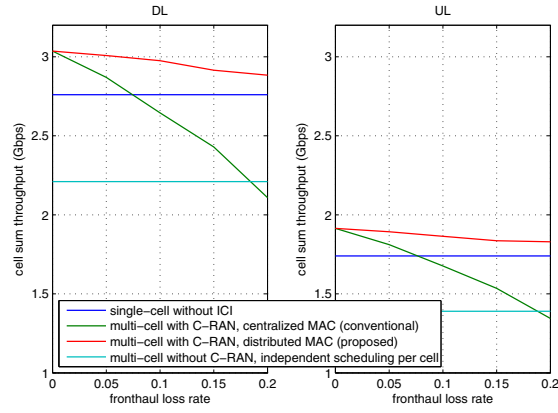


Fig. 7. Simulation result: DL and UL cell sum throughput with different fronthaul loss rates.

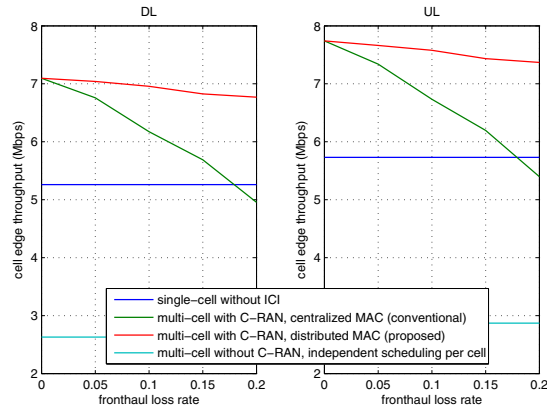


Fig. 8. Simulation result: DL and UL cell edge throughput with different fronthaul loss rates.

due to a large spatial multiplexing degree in massive MIMO. If C-RAN is utilized (case 2 and 3), the cell sum throughput is slightly increased but the cell edge throughput is significantly boosted, mainly because of the effect of CoMP. The case of multi-cell without C-RAN (case 4) has the worse throughput due to severe ICI. Then, with non-ideal fronthaul, when the loss rate is increased, the benefit of distributed MAC scheduling becomes obvious. The performance of conventional centralized MAC scheduling deteriorates dramatically, which is almost linearly with the loss rate, since no special handling is taken for the lost fronthaul messages of MAC control, including scheduling grants, HARQ feedback and CSI. On the opposite, with the proposed distributed MAC scheduling scheme, the performance loss due to non-ideal fronthaul could be compensated in a large degree by the limited scheduling at RAU. Specifically, the cell sum throughput performance degradation by non-ideal fronthaul with 5%, 10%, 15%, 20% loss rate is around 1%, 2%, 4%, 6% by the proposal distributed scheme, while it is around 5%, 13%, 20%, 30% by the conventional centralized scheme.

V. CONCLUSION

This paper proposes a distributed MAC scheduling scheme for C-RAN with non-ideal fronthaul, involving a full scheduling operation at RCC and a limited scheduling operation at RAU. Typical approaches of the full scheduling and the limited scheduling together with their benefits are illustrated. In addition, the integration of these two scheduling results are also provided when they are both available at RAU. It is demonstrated by simulation results that the proposed scheme could effectively compensate the cell throughput loss caused by non-ideal fronthaul, and hence promote C-RAN feasibility in 5G cellular networks.

REFERENCES

- [1] Recommendation ITU-R M.2083, "IMT Vision - Framework and overall objectives of the future development of IMT for 2020 and beyond," Sep. 2015.
- [2] METIS II white paper, "5G RAN Architecture and Functional Design," Mar. 2016.
- [3] Chih-Lin I et al., "Toward Green and Soft: A 5G Perspective," *IEEE Communications Magazine*, vol. 52, no. 2, Feb. 2014, pp. 66-73.
- [4] Chih-Lin I et al., "Recent Progress on C-RAN Centralization and Cloudification," *IEEE Access*, vol. 2, 2014, pp. 1030-1039.
- [5] Sheng Zhou et al., "Software-Defined Hyper-Cellular Architecture for Green and Elastic Wireless Access," *IEEE Communications Magazine*, vol. 54, no. 1, Jan. 2016, pp. 12-19.
- [6] Chih-Lin I et al., "Rethink Fronthaul for Soft RAN," *IEEE Communications Magazine*, vol. 53, no. 9, Sep. 2015, pp. 82-88.
- [7] Zhiyong Feng et al., "An Effective Approach to 5G: Wireless Network Virtualization," *IEEE Communications Magazine*, vol. 53, no. 12, Sep. 2015, pp. 53-59.
- [8] Matteo Fiorani et al., "Energy Performance of C-RAN with 5G-NX Radio Networks and Optical Transport," in *Proc. of IEEE ICC*, 2016.
- [9] Chia-Yu Chang et al., "Impact of Packetization and Functional Split on C-RAN Fronthaul Performance," in *Proc. of IEEE ICC*, 2016.
- [10] 3GPP Technical Report 38.801, "Study on New Radio Access Technology; Radio Access Architecture and Interfaces (Release 14)," 2016.
- [11] Jian Zhao et al., "Fast Algorithm for Utility Maximization in C-RAN with Joint QoS and Fronthaul Rate Constraints," in *Proc. of IEEE ICC*, 2016.
- [12] Seung Hwan Kim et al., "Experimental Demonstration of CPRI Data Compression based on Partial Bit Sampling for Mobile Front-Haul Link in C-RAN," in *Proc. of Optical Fiber Communications Conference and Exhibition (OFC)*, 2016.
- [13] Wonju Lee et al., "Multivariate Fronthaul Quantization for C-RAN Downlink: Channel-Adaptive Joint Quantization in the Cloud," in *Proc. of IEEE ICC*, 2016.
- [14] Wonju Lee et al., "Multivariate Fronthaul Quantization for Downlink C-RAN," *IEEE Trans. Signal Processing*, vol. 64, no. 19, Oct. 2016, pp. 5025-5037.
- [15] Chih-Lin I et al., "White paper of next generation fronthaul interface (v1.0)," China Mobile Research Institute, Alcatel-Lucent, Nokia Networks, ZTE Corporation, Broadcom Corporation, Intel China Research Center, *Tech. Rep.*, 2015.
- [16] Wei Feng et al., "Virtual MIMO in Multi-Cell Distributed Antenna Systems: Coordinated Transmissions with Large-Scale CSIT," *IEEE Journal on Selected Areas in Communications*, vol. 31, no. 10, Oct. 2013, pp. 2067-2081.
- [17] Lifeng Wang et al., "Secrecy and Energy Efficiency in Massive MIMO Aided Heterogeneous C-RAN: A New Look at Interference," *IEEE Journal of Selected Topics in Signal Processing*, vol. 10, no. 8, Dec. 2016, pp. 1375-1389.
- [18] 3GPP Technical Specification 36.300, "E-UTRAN Overall Description (Release 13)," 2016.
- [19] 3GPP Technical Report 36.873, "Study on 3D channel model for LTE (Release 12)," 2016.
- [20] TS V5G.211, "Physical Channels and Modulation," *Verizon 5G Technical Forum*, <http://www.5gtf.org/>.